

HUTCHINSON, MN, USA

WMO: 726569

Lat: 44.859N

Lon: 94.382W

Elev: 323

StdP: 97.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.8	-29.7	0.3	-25.4	-26.8	0.3	-22.8	12.6	-11.5	11.6	-10.1	2.7	300	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.2	23.9	30.2	22.6	28.5	21.7	25.6	29.9	24.3	28.2	23.1	26.8	4.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	28.4	22.9	18.4	26.8	22.1	17.4	25.8	80.5	29.8	74.9	28.3	70.4	26.7	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.1	9.4	8.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-29.1	35.0	3.8	1.6	-31.8	36.1	-34.0	37.1	-36.1	37.9	-38.8	39.1		
			-29.3	27.6	3.5	1.2	-31.8	28.5	-33.9	29.2	-35.9	29.9	-38.4	30.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-9.9	-8.0	-1.0	7.3	14.1	19.8	22.0	20.4	16.5	8.9	0.8	-6.4
	DBStd	12.49	7.51	7.25	6.98	5.45	4.86	3.72	3.25	2.99	4.64	5.17	5.82	6.64
	HDD10.0	2473	617	503	344	113	16	0	0	0	5	84	281	509
	HDD18.3	4417	876	737	598	332	150	26	6	13	89	296	527	768
	CDD10.0	1422	0	0	4	31	142	296	373	322	200	49	4	0
	CDD18.3	324	0	0	0	1	18	71	121	77	34	3	0	0
	CDH23.3	2934	0	0	1	23	212	634	1097	619	306	41	1	0
	CDH26.7	851	0	0	0	6	69	198	348	143	79	8	0	0
Wind	WSAvg	3.6	4.0	3.9	4.2	4.5	4.2	3.5	2.7	2.4	3.0	3.7	4.0	3.7
Precipitation	PrecAvg	762	16	16	39	68	94	121	96	105	78	58	46	19
	PrecMax	995	37	29	98	179	176	230	206	193	177	151	129	53
	PrecMin	528	1	1	6	12	28	63	33	11	11	10	3	6
	PrecStd	125	10	9	23	36	39	44	46	46	45	39	38	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	10.4	20.0	26.3	32.3	33.9	34.7	32.4	32.1	27.5	19.0	10.8
		MCWB	3.4	8.3	14.1	15.7	19.9	23.1	26.6	25.2	22.8	17.3	12.8	7.8
	2%	DB	2.9	5.7	15.1	22.5	28.2	31.2	32.3	29.9	28.8	23.7	14.9	6.3
		MCWB	1.7	3.6	11.9	13.2	17.5	21.9	25.3	24.1	21.1	16.1	9.9	3.7
	5%	DB	1.4	3.1	12.0	19.4	25.4	28.8	30.3	28.0	27.1	20.7	12.0	3.6
		MCWB	0.6	1.9	9.7	12.0	16.8	20.5	23.9	22.3	19.7	14.8	8.3	2.1
	10%	DB	0.2	2.0	8.0	16.3	22.7	27.1	28.3	27.0	24.8	17.3	8.9	2.0
		MCWB	-0.4	1.0	5.9	10.0	15.3	19.4	22.4	21.3	19.0	12.5	5.9	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	8.5	16.2	17.1	22.2	25.4	28.0	26.4	24.7	20.2	14.6	8.2
		MCDB	5.7	10.0	18.0	23.3	29.0	30.7	32.4	30.6	28.9	24.1	18.2	9.4
	2%	WB	1.8	3.8	12.1	14.7	20.1	23.8	26.3	24.9	22.8	17.4	10.7	3.8
		MCDB	3.0	5.3	14.3	20.3	24.8	28.4	30.7	28.7	26.4	21.5	14.4	5.7
	5%	WB	0.7	2.2	9.0	12.8	18.4	22.2	24.9	23.6	21.5	15.2	8.1	2.2
		MCDB	1.3	3.2	11.8	17.7	22.9	26.8	28.9	27.0	25.2	19.3	11.1	3.6
	10%	WB	-0.5	0.9	5.9	11.0	16.7	20.9	23.6	22.4	19.9	13.1	6.1	0.9
		MCDB	0.1	1.6	8.2	15.8	21.1	25.1	27.4	25.5	23.6	16.9	8.9	1.7
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.7	9.1	11.0	11.5	10.9	11.0	11.1	12.3	11.2	9.3	8.5
		MCDBR	9.1	10.0	13.3	16.6	16.1	13.7	12.0	12.3	14.4	15.9	14.2	11.2
		MCWBR	8.1	8.3	9.4	9.5	7.9	6.4	6.6	7.0	7.5	8.9	9.6	8.8
	5% WB	MCDBR	8.8	9.1	12.9	14.3	13.1	11.2	11.1	11.1	12.0	13.2	12.4	10.2
		MCWBR	7.9	7.6	9.5	9.2	7.7	6.4	6.7	7.0	7.3	8.2	8.9	8.3
Clear-Sky Solar Irradiance	taub	0.258	0.280	0.307	0.361	0.389	0.394	0.399	0.405	0.368	0.317	0.283	0.260	
	taud	2.456	2.388	2.434	2.341	2.318	2.354	2.341	2.307	2.383	2.525	2.527	2.464	
	Ebn at Noon	879	915	932	900	880	875	866	845	851	862	841	839	
	Edh at Noon	69	89	98	118	125	121	121	121	103	78	65	62	
All-Sky Solar Radiation	RadAvg	1.78	2.83	3.75	4.64	5.38	5.99	6.44	5.34	4.17	2.69	1.73	1.28	
	RadStd	0.21	0.32	0.36	0.45	0.42	0.46	0.23	0.42	0.31	0.36	0.16	0.14	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page